

(allow reasonable alternatives)

- 3 (a) (i) ductile B1
- (ii)1 L shown at end of straight line B1
- (ii)2 reciprocal of gradient of straight line region B1 [3]
- (b) (i)1 circumference = 3 cm or arc = $r\theta$ C1
extension = $(6.5/360) \times 3\pi$ = 1.5 sin (or tan) 6.5.....M1
= 0.17 cm A0
- (i)2 strain = extension/length..... C1
= 0.17/250
= 6.8×10^{-4} A1 [4]
- (ii) stress = force/area C1
= $(6.0 \times 9.8)/(7.9 \times 10^{-7})$ C1
= 7.44×10^7 Pa A1 [3]

